

Day 36 - Lagrangian Examples III

WHEN YOU START TO LEARN LAGRANGIAN MECHANICS



Announcements

- Last "Class" Week
- Homework 8 due Friday, Nov 21st (late after Nov 26th)
- Next Week: Project Work and Discussion
- Last Week: Presentations
- Final Project Due Dec 8th (no later than 11:59 pm)
- **No Final Exam**



Complete Google Form

By November 21st

Reporting your group members for the final project and a short summary of your project idea for sharing with the class.

<https://forms.gle/iPKR9EDAaHW3GirN7>

Reminders

We found the Lagrangian for the Atwood's machine with a rotating pulley to be:

$$\mathcal{L} = \frac{1}{2}(M + m)R^2\dot{\phi}^2 + \frac{1}{4}M_pR^2\dot{\phi}^2 - (M - m)gR\phi$$

where M is the mass of the left block, m is the mass of the right block, M_p is the mass of the pulley, R is the radius of the pulley, and ϕ is the angle of rotation of the pulley.

We used the scleronomic constraint $y_1 = R\phi$ to do this.

Clicker Question 36-1

We derived the Lagrangian for the Atwood's machine with a rotating pulley to be:

$$\mathcal{L} = \frac{1}{2}(M + m)R^2\dot{\phi}^2 + \frac{1}{4}M_pR^2\dot{\phi}^2 - (M - m)gR\phi$$

What is generalized force, $F_\phi = \partial\mathcal{L}/\partial\dot{\phi}$?

1. $+(M - m)gR$
2. $-(M - m)gR$
3. $+(M + m)R^2\dot{\phi}$
4. $-(M + m)R^2\dot{\phi}$
5. Something else

Clicker Question 36-2

We derived the Lagrangian for the Atwood's machine with a rotating pulley to be:

$$\mathcal{L} = \frac{1}{2}(M + m)R^2\dot{\phi}^2 + \frac{1}{4}M_pR^2\dot{\phi}^2 - (M - m)gR\phi$$

What is the generalized momentum, $p_\phi = \partial\mathcal{L}/\partial\dot{\phi}$?

1. $+(M - m)gR$
2. $-(M - m)gR$
3. $+(M + m)R^2\dot{\phi}$
4. $-(M + m)R^2\dot{\phi}$
5. Something else

Clicker Question 36-3

For the constraint for the bead in a parabolic bowl ($z = c\rho^2$), what are the units of c ?

1. $[L^2]$
2. $[L^{-2}]$
3. $[L]$
4. $[L^{-1}]$
5. Something else

Clicker Question 36-4

For the bead in a parabolic bowl, there is a generic Lagrangian:

$$\mathcal{L}(\rho, \dot{\rho}, \phi, \dot{\phi}, z, \dot{z}, t)$$

How many coordinates are there, truly? **here, each variable is a coordinate**

- A. 2
- B. 3
- C. 4
- D. 5
- E. None of these

Which coordinates are independent?

Clicker Question 36-5

The Lagrangian for the bead in a parabola does not depend on which of the following?

1. ρ
2. ϕ
3. z
4. More than one of these
5. None of these